Financing the Korean State, 1961–2023:

The Shifting Burden of Taxes, Bonds, and the Inflation Tax*

Hyungsuk Lee[†]
Bank of Korea

Hyun Ji Lee[‡]
Bank of Korea

June 2026

Abstract

We apply the government budget-constraint accounting of Hall and Sargent (1997, 2011, 2021, 2022) to reconstruct how Korea financed its government from 1961 to 2023, and who bore the cost. Through the high-inflation development decades the state ran large deficits yet kept its debt low—not by liquidating bonds, as the advanced economies did, but through rapid growth and an inflation tax that fell on holders of *money*: the bond market was still too small for inflation to tax bondholders, so the levy worked through seigniorage worth nearly two percent of GDP a year. As Korea built a sovereign bond market and an inflation target, the financing of its fiscal pressures, and their incidence, migrated from money to bonds to taxes—the Korean War paid for by the printing press, the 1997 crisis off the books through state guarantees, COVID by conventional bonds. The same accounting, pointed forward, shows the inflation-tax engine retired: an aging Korea's deficits must now be paid in taxes, not inflated away. The progression distinguishes a developing-country fiscal history from the bond-financed pattern of the United States.

JEL Classification: E31, E62, E63, H63, N15

Key Words: government budget constraint, inflation tax, seigniorage, fiscal capacity, sovereign bond market, public debt, Korea

*The views expressed here are those of the author and do not necessarily reflect the official positions of the Bank of Korea. This is a preliminary draft; comments welcome.

†Economic Research Institute, Bank of Korea, 39 Namdaemun-ro, Jung-gu, Seoul, Korea; email: walden0230@gmail.com.

‡Junior Economist, Bank of Korea; email: lhj@bok.or.kr.

1 Introduction

This paper reconstructs how the Korean state financed itself across seven decades and three great shocks, and finds that the burden of its fiscal pressures migrated from money to bonds to taxes as the country built a sovereign bond market. We take the government budget-constraint accounting of Hall and Sargent (1997, 2011, 2021, 2022)—which reconstructs the market value of public debt and reads the financing of wars and crises off the government's intertemporal budget constraint—to Korea from 1961 to 2023, to its three largest fiscal shocks (the Korean War, the 1997 Asian financial crisis, and COVID-19), and to a projection of its aging future. Applied to the United States, that framework describes a government that paid for its wars by issuing marketable debt and repaid bondholders in part through subsequent inflation. Korea tells a different story, because for most of this history it had no deep bond market to issue into: each shock was met in the dominant financing mode its institutions then allowed—the inflation tax in 1950, off-budget guarantees in 1997, conventional bonds and taxes in 2020—and the incidence of fiscal pressure shifted with the instruments the state had built.

Korea held its public debt remarkably flat for three decades not by liquidating a stock of bonds—the post-war financial-repression channel of Reinhart and Sbrancia (2015)—but through an inflation tax that fell on money. From 1961 to the mid-1980s the debt ratio barely moved even as the government ran persistent deficits at double-digit nominal interest: rapid real growth and high inflation eroded the real value of the debt about as fast as borrowing added to it, and the inflation tax was collected as seigniorage on base money—on the order of two percent of GDP a year—rather than as a capital levy on bondholders. It fell on money rather than bonds for a simple institutional reason: Korea had barely any marketable bonds to tax. Through the development decades the traded government bond stock was on the order of three percent of GDP—about a fifth of a public debt that was itself small and dominated by foreign borrowing and captive domestic loans—and the 1965 interest-rate reform deliberately held deposit and lending rates above inflation, so the real return on debt was rarely negative. This is not the Reinhart–Sbrancia world of a large bond stock slowly liquidated by sustained negative real rates; it is an inflation tax on the holders of money, levied by a state that had not yet built the bond market through which the textbook channel runs.

The three great shocks are where this logic is sharpest, because each was paid for in the dominant mode its era allowed. The Korean War (1950–53) was financed by the printing press: with essentially no marketable debt and a collapsed tax base, the government covered much of its spending through note issue, and wholesale prices rose roughly eighteenfold between 1950 and 1953—

an inflation tax extracted almost entirely from holders of money. The 1997 crisis was financed off the books: a state-led financial system absorbed the cost through government-guaranteed KAMCO and KDIC bonds—contingent liabilities that standard debt statistics miss until they are called—which later surfaced on the national accounts and explain much of the debt's post-1997 climb. COVID-19 was financed conventionally: by 2020 a deep Treasury market spanning maturities from one to fifty years let Korea borrow as an advanced economy does, and the inflation and rate increases of 2021–22 imposed a capital loss on bondholders—the very revaluation channel that had no purchase when the bonds did not yet exist. The thread is a single point in the spirit of Lucas and Stokey (1983): a government can inflate away only the debt it has issued, so as Korea built a bond market the financing of its shocks, and the people who bore it, moved from money-holders toward bondholders and taxpayers.

This history carries a sharp implication for Korea's future, which we trace with the same accounting. The capacity that gave Korea a deep bond market and a credible inflation target also retired the inflation-tax engine: a state that has promised two percent inflation, and whose debt now earns a market return close to its growth rate, can no longer let growth and surprise inflation quietly erode its deficits. Applying the budget-constraint identity to the long-run projection of the National Assembly Budget Office, we find a debt ratio that more than triples—from 48 percent of GDP in 2025 to 173 percent in 2072—driven almost entirely by the primary deficits that population aging opens up rather than by any interest-rate snowball: with the Treasury rate sitting near nominal growth, the growth-adjusted interest term stays near zero, and seigniorage delivers about 0.2 percent of GDP a year against the nearly two percent it yielded in the 1970s. The forces that held the developmental debt flat are gone, so the aging-driven deficits ahead accumulate point for point into debt that can be closed only through taxes and spending. The escape valve that paid for Korea's first great shock was sealed by the very institutions that mark its arrival as a mature fiscal state.

Our contribution is fourfold. First, we carry the Hall–Sargent budget-constraint accounting—both its continuous debt/GDP decomposition and its shock-financing decomposition—to a developing economy across its entire passage from monetary to market finance, whereas prior applications of the framework are almost all to advanced economies, and to the United States above all. Second, we establish that Korea's development era does not fit the financial-repression mold of Reinhart and Sbrancia (2015): the inflation tax fell on money through seigniorage, not on bondholders through liquidation, and we *measure* the incidence—who, across seven decades, actually paid for the Korean state—rather than asserting it. Third, we bring three very different shocks—a war with no usable national accounts, a banking crisis financed off the books, and a pandemic financed in a deep bond market—into a single budget-constraint framework, reconstructing the Korean War from primary sources in levels and financing shares where no won-denominated GDP exists. Fourth, we point the same identity forward, decomposing an official projection of an aging Korea's budget to show that the inflation-tax engine, having done so much work in the past, is no longer available for the deficits to come.

We measure all of this by reconstructing Korea's consolidated government budget constraint from primary Korean sources. Consolidating the Bank of Korea with the Treasury is essential in a developing economy, where much of what the accounts label "borrowing" was direct central-bank lending that the consolidation correctly records as money creation rather than market debt. Our series are built from the long-run statistics of the Naksungdae Institute (debt, revenue, and expenditure from the 1950s), the Bank of Korea's Economic Statistics System (money, prices, and the full term structure of bond yields), and the Bank's wartime fiscal–monetary compendium, spliced to official accounts from 2016; we work on a central-government basis, take debt as concept D1 net of government guarantees, and keep those guarantees as a separate contingent series—the off-budget channel through which the 1997 crisis was financed. We read the budget constraint in two complementary ways: a law of motion for the debt/GDP ratio, applied to every year, which splits the annual change into the contributions of the primary deficit, interest, real growth, inflation, and seigniorage; and a shock-financing decomposition, applied to each episode, which allocates a spending surge across taxes, new borrowing, money creation, and the erosion of outstanding debt by growth and inflation. We are explicit about the two places the data bind us—the splices between sources, and the absence of market bond prices before about 2000, for which we use an administered-rate proxy whose error the small early bond stock contains.

The accounting yields a single arc. Averaged over 1961 to 2023 the debt ratio rose by barely half a point of GDP a year, but that calm is the sum of a rise and a fall. Through the development decades growth, inflation, and seigniorage eroded the debt about as fast as deficits and double-digit interest added to it, holding the ratio near where it began; in the decade before 1997, with the budget in surplus, the same forces pushed it down. The engine then died: around 1997 inflation and seigniorage fell to near zero just as the off-budget crisis debt surfaced on the books—a stock–flow adjustment of some 2.7 points of GDP a year over 1997–2007—and the debt began a climb it has not reversed. COVID completed the inversion, the primary deficit alone driving the ratio up nearly three points a year, the fastest in the record. Read as a question of incidence, the burden moved from the holders of money in the development era, through the captive and contingent creditors of the crises, to the taxpayers and bondholders of the present—and, on the projection, to the taxpayers of the future.

This paper sits at the intersection of five literatures.

The first is the government budget-constraint accounting of Hall and Sargent, who reconstruct the market value of public debt maturity by maturity and read the financing of wars and crises off the intertemporal budget constraint (Hall and Sargent, 1997, 2011, 2014, 2021, 2022, 2023, 2025). Their evidence is overwhelmingly American, and their two tools have largely been used apart: a continuous decomposition of post-war debt/GDP dynamics in Hall and Sargent (2011), and a shock-by-shock financing decomposition of wars in Hall and Sargent (2021, 2022). We carry both to a developing economy and join them in one accounting, from an inflation-financed war to a COVID financed in a deep bond market. The framework's central measurement—the holding-period return on debt, and the capital loss that inflation and rising rates impose on bondholders—has a direct empirical descendant in Hilscher et al. (2022), who price the debasement of nominal debt maturity by maturity and find it small for the modern United States but large under financial repression. That mechanism is exactly the one through which Korea's compulsory development-era bonds, and the rate spike of 2022, bear on the incidence we measure.

The second is the economics of fiscal and state capacity. Besley and Persson (2009, 2011) and Dincecco (2011) treat the ability to tax and to borrow as capital a state accumulates over time. Korea's migration from the inflation tax to marketable debt is exactly such an accumulation, measured here through the budget constraint; and the bond market it built is an escape from what Eichengreen and Hausmann (1999) and Eichengreen et al. (2005) call original sin—the historical inability of developing countries to borrow long-term in their own currency. Korea, with no marketable long-term won bond market before 2000 and now issuing to fifty years and selling a fifth of it abroad, is a clean case of that escape.

The third is the inflation tax, seigniorage, and financial repression. Reinhart and Sbrancia (2015) and Reinhart et al. (2011) show how negative real rates liquidated the large debts of the post-war advanced economies; Sargent and Wallace (1981) formalize the deficit–inflation link, Cagan (1956) and Click (1998) the revenue from money creation, and Reis (2021) the distinct "debt revenue" a government earns when its bonds pay below the marginal product of capital—a wedge that, under coercion, becomes the repression premium visible in Korea's captive placements. The structural counterpart of the unpleasant arithmetic has been revisited by Uribe (2017) and Werning (2024), who recast it as a seigniorage Laffer curve and an inflation-tax-smoothing problem; we take their lesson as a caveat—seigniorage is an intertemporal, present-value object, so realized period-by-period incidence must be read against that benchmark—while keeping our own measurement reduced-form. This margin has been surveyed for war finance and the long deficit–inflation record by Bordo and Levy (2021), in which Korea figures, if at all, as one case among many rather than the subject. A central—and to us surprising—finding is that Korea's development era does *not* fit the Reinhart–Sbrancia mold: the 1965 interest-rate reform kept real rates positive, so the inflation tax fell on money through seigniorage, not on a bond market the country lacked.

The fourth is the accounting of debt dynamics and off-budget liabilities. We use the stock–flow framework of Escolano (2010) and Abbas et al. (2011), and—for 1997—the literature on contingent government liabilities (Polackova, 1998), which names the very mode of financing that standard debt statistics miss: state guarantees that become debt only when called.

The fifth is the fiscal and monetary history of Korea. We build on the long-run reconstructions of Kim (2012); Kim et al. (2018) and the monetary history of Kim and Lee (2010). The closest prior work for Korea is Koh (2005), who applies the fiscal-risk and contingent-liability framework of Polackova (1998) to assess Korea's fiscal *sustainability* and exposure prospectively—including the state guarantees that loom large in our 1997 episode. We use the same contingent-liability lens to a different end: to *measure*, within a budget-constraint accounting, how past shocks were actually financed and how that mode shifted. Finally, we connect to the consolidated, generational view of Korean public finance in Chien et al. (2025), of which our projection of an aging Korea's budget constraint (Section 5) is a natural complement.

The paper proceeds as follows. Section 2 sets out the accounting framework and the data. Section 3 reads Korea's continuous record from 1961 to 2023, locating the rise and fall of the inflation-tax engine and the incidence it implies. Section 4 turns to the three shocks—the Korean War, 1997, and COVID-19—each financed in the dominant mode of its era. Section 5 points the same identity forward, decomposing the projected budget of an aging Korea. Section 6 concludes.

2 Accounting framework and data

This section sets out the tools and the evidence in turn. Section 2.1 lays out the accounting—the consolidated government budget constraint and the two complementary ways we read it: a year-by-year law of motion for the debt ratio, which drives the continuous record of Section 3, and a shock-financing decomposition, which we take to the episodes of Section 4. Section 2.2 then describes how each term of that accounting is measured from Korean primary sources across seven decades, and is candid about the two places—the splices between sources and the late birth of a bond market—where our precision is bounded.

2.1 The accounting framework

Our accounting rests on the *consolidated* flow budget constraint of the government—the Treasury together with the central bank—which we write following Hall and Sargent (2021, 2022) and, ultimately, Sargent and Wallace (1981). Let B_t be the end-of-period nominal value of interest-bearing government debt held *outside* the public sector, $r^B_{t-1,t}$ the value-weighted nominal holding-period return on it, G_t outlays net of interest (purchases and transfers), T_t taxes net of transfers, and M_t the monetary base. Uses of funds equal sources:

$$G_t + r^B_{t-1,t} B_{t-1} \;=\; T_t + (B_t - B_{t-1}) + (M_t - M_{t-1}) + A_t, \tag{1}$$

Consolidating the Bank of Korea with the Treasury is what gives (1) its bite in a developing economy: through the developmental decades, much of what the accounts label "borrowing" was direct central-bank lending to the government, which the consolidation correctly records as money creation rather than market debt. Two conventions, both from Hall and Sargent (1997, 2011), govern the constraint: B is held outside the public sector (net of central-bank and government-account holdings) and is valued at *market*, not par, prices wherever a market exists—for the pre-2000 period, where none did, we use par. The external term A_t—grants and advances, negligible for most of the sample but central to the Korean War (Section 4.1) and the counterpart of Hall and Sargent (2019)'s foreign-credit term—we take up next.

The external term deserves a sharper treatment, because two of our episodes turn on resources whose value was uncertain when they were extended. We follow the foreign-credit accounting of Hall and Sargent (2019), in which a liability backed by a claim of uncertain value is netted against that claim: the relevant object is $B_t - C_t$, where C_t is the present value of the backing asset, and the *ex post* fiscal cost is the gross figure less what the asset ultimately delivers. Two Korean cases share this structure. During the Korean War, U.S. grants enter (1) as an external non-tax source (A_t), while repayable U.S. and UN advances form an external liability backed by later won–dollar settlement. In 1997, the government's guaranteed KAMCO and KDIC bonds were claims backed by recoveries on impaired bank assets, so the headline injection overstates the cost, which is the un-recovered remainder. Treating both under the single object $B - C$ keeps the war's aid and the crisis's contingent debt commensurable and prevents gross flows from masquerading as the true burden.

Setting A_t aside (negligible over 1961–2023, and entering the Korean War only in levels) and dividing by nominal GDP Y_t, the identity admits two complementary representations that we use throughout. The first, applied *every* year, is a law of motion for the debt/GDP ratio:

$$\frac{B_t}{Y_t} - \frac{B_{t-1}}{Y_{t-1}} = r_{t-1,t}\frac{B_{t-1}}{Y_{t-1}} - g_{t-1,t}\frac{B_{t-1}}{Y_{t-1}} - \pi_{t-1,t}\frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t} - \frac{M_t - M_{t-1}}{Y_t} + \varepsilon_t, \tag{2}$$

which splits each year's change in debt/GDP into the contributions of the primary deficit, the return on debt, real growth, inflation, and seigniorage, plus a residual ε_t (the stock–flow adjustment, discussed in Section 3). We apply (2) annually over 1961–2023—the war years 1950–52 lack won-denominated GDP and are handled in levels (Section 4.1)—and read this single identity across two regimes, one in which the inflation tax did the work and one of market finance. The second representation, obtained by summing deviations from a pre-shock "peacetime baseline" over the shock years $t = T_1, \ldots, T_2$, is the financing decomposition

$$\begin{aligned}\underbrace{\textstyle\sum_t \Delta \frac{G_t}{Y_t}}_{\text{spending surge}} + \underbrace{\textstyle\sum_t \Delta\left(r_{t-1,t}\frac{B_{t-1}}{Y_{t-1}}\right)}_{\text{return on debt}} = \underbrace{\textstyle\sum_t \Delta \frac{T_t}{Y_t}}_{\text{taxes}} + \underbrace{\textstyle\sum_t \Delta\left(\frac{B_t}{Y_t} - \frac{B_{t-1}}{Y_{t-1}}\right)}_{\text{borrowing}} + \underbrace{\textstyle\sum_t \Delta \frac{M_t - M_{t-1}}{Y_t}}_{\text{seigniorage}} \\ + \underbrace{\textstyle\sum_t \Delta\left(g_{t-1,t}\frac{B_{t-1}}{Y_{t-1}}\right)}_{\text{growth dilution}} + \underbrace{\textstyle\sum_t \Delta\left(\pi_{t-1,t}\frac{B_{t-1}}{Y_{t-1}}\right)}_{\text{inflation dilution}} + (\text{cross term}),\end{aligned} \tag{3}$$

where g is real GDP growth and π inflation. Equation (3) is our central tool: it allocates a spending surge across explicit taxes, new marketable borrowing, money creation, and the erosion of the real value of outstanding debt by growth and inflation.

The return r^B in (1) is not the coupon the Treasury reports as "interest" but the value-weighted holding-period return actually earned by those who hold the debt—the object Hall and Sargent (1997) identify as the government's true cost of funds. It has two parts. The *explicit* part is the coupon and principal paid out; the *implicit* part is the capital gain or loss the market records as yields move—when interest rates or inflation rise, the price of outstanding long bonds falls, the government's creditors lose, and its real cost of funds falls correspondingly. Value-weighting the holding-period return across outstanding maturities delivers r^B; the implicit term is exactly what official interest-expense series omit, and its variability is the "interest-rate risk" of Hall and Sargent (2011). The distinction is not academic for Korea: it is the channel through which COVID-era inflation and rate increases imposed a large real loss on bondholders (Section 4.4). Measuring it requires market prices, which exist only from about 2000; for earlier years we approximate r^B by a single market yield and treat the resulting gap as a measurement caveat.

A key conceptual point organizes the Korean results. Inflationary finance operates through *two* channels: (i) *seigniorage*, the inflation tax on real money balances (the $\Delta(M_t - M_{t-1})/Y_t$ term), which requires no bond market; and (ii) *debt revaluation*, the capital levy that surprise inflation imposes on holders of nominal bonds (the $\pi_{t-1,t}B_{t-1}/Y_{t-1}$ term), which requires a stock of nominal debt. Where the debt stock is negligible—as in Korea in 1950—inflationary finance can work only through channel (i). This is a Lucas–Stokey point: a government can inflate away only the nominal debt it has actually issued. The two channels therefore map onto the instruments a state has built—the inflation tax on money is always available, but the inflation tax on bondholders presupposes a bond market. Korea's three shocks span exactly that transition, which is why the same identity lights up different terms in each.

Each term of (1), finally, names not only a source of funds but a party that gives them up—which is how we read the decomposition throughout. Taxes fall on current taxpayers; new marketable borrowing is, in present value, a claim on future ones. Seigniorage is the inflation tax on holders of real money balances, and the inflation-dilution term πB is the same tax levied on holders of nominal bonds, whose real claims it silently reduces; the realized return r^B distributes interest-rate risk between the government and its creditors as bond prices move. Forced placements—bonds a captive class is compelled to hold below market terms—are a further levy, falling on whoever cannot refuse them. This is the language of fiscal discrimination in Hall and Sargent (2014): a government can shift the cost of its spending across classes of claimants, and which class bears it is itself a policy outcome (cf. Reinhart and Sbrancia, 2015, on financial repression). Our organizing question is therefore one of incidence—across seven decades, *who* has paid for the Korean state, and how the answer has changed.

2.2 Data and measurement

Our measurement is built, wherever possible, from Korean primary sources, consolidating the Bank of Korea with the Treasury as the framework requires. Table 1 lists the series and their coverage. The defining difficulty is reach: the accounting asks for debt, fiscal flows, money, prices, and yields across seven decades that contain a war, a developmental transformation, two currency reforms, and the late birth of a bond market. No single source spans them, so we splice—and we are explicit about where the splices and the absence of historical market prices bound our precision. Construction details are in Appendix A.

Government debt is the series that most shapes our results, and the one where a Korean primary source is decisive. We take central-government debt from the long-run statistics of the Naksungdae Institute (Kim et al., 2018), which reconstruct it annually from 1948 and decompose it into marketable bonds, loans (domestic and foreign), and government-guaranteed liabilities. Our debt measure B is central-government debt excluding guarantees, which tracks the official D1 series to within one or two points of GDP over their post-1990 overlap—the small gap being local-government debt. We keep government-*guaranteed* debt as a separate, contingent series rather than folding it into B: it is the off-budget channel through which the 1997 crisis was financed (Section 4.2), reaching some 11% of GDP around 2000, and it also captures the developmental state's guarantees on foreign borrowing. Marketable bonds were a small fraction of the total before about 2000. From 2016 we splice to the official D1 series; for a modern general-government view (including the National Pension Service) we report D2 as a supplement.

Government revenue and expenditure—the flows T and G of the budget constraint—come from the same Naksungdae source on a central-government basis, annually from 1954, with the war years from Bank of Korea (2005). The Naksungdae series usefully reports the fiscal balance both gross and *net of foreign aid*, which lets us isolate the aid term A_t of equation (1)—decisive for the Korean War, when aid covered nearly a quarter of spending (Section 4.1). Consistent with the debt measure, G is outlays net of interest and T is revenue net of transfers; the primary deficit $G - T$ is the object that enters the decomposition. From 2016 we use the official managed-balance and consolidated central-government accounts of the Ministry of Economy and Finance.

Money and prices are the cleanest of our long-run series. The monetary base M—the non-interest-bearing liability of the consolidated government, whose change is seigniorage—is continuous from 1960 (Bank of Korea, ECOS), extended through the war by the currency-issue figures in Bank of Korea (2005). For prices we use the Bank of Korea's producer price index, which runs on a single, consistent concept from 1910 to the present (the historical series 404Y017 chained to its modern continuation) and so spans the Korean War without a definitional break; its monthly version captures the wartime hyperinflation directly, with wholesale prices rising roughly eighteenfold between 1950 and 1953. Inflation π in the decomposition is the producer-price change; consumer prices, available from 1965, give very similar results in the modern period.

The return on debt is the one input we cannot measure cleanly throughout, for a simple reason: Korea had no market in long-term government bonds before about 2000, so no market yield exists to price the early debt. We therefore splice. For 1945 to the late 1990s we use the Bank of Korea's lending rate (Bank of Korea, 2005)—an administered rate, but the relevant cost of funds when borrowing was largely from the central bank; from 1987 the Naksungdae statistics add a government (housing) bond yield as a cross-check; and from the late 1990s we use the full term structure of KTB yields (ECOS), which is what the market-value return r^B of Section 2 requires. We are explicit that the early portion is a proxy, but two facts limit the damage: in the developmental decades the debt stock B is small, so the term $r^B B$ that the rate multiplies is small; and where B is large—the modern period—a genuine market yield is available. The residual this approximation leaves is taken up with the decomposition (Section 3).

The exchange rate and the national accounts come from ECOS, with war-era figures from Bank of Korea (2005), the compendium underlying Kim and Lee (2010).

One gap is irreducible. Won-denominated GDP does not exist for 1941–52: the war and the colonial-to-independence transition destroyed the statistical basis, and the period is left blank even in the most complete long-run reconstruction (Kim, 2012), which splices the colonial accounts (to 1940) to the post-war Bank of Korea series (from 1953). No source can fill it. This is why the continuous decomposition of Section 3 begins only in the mid-1950s, and why the Korean War is accounted in levels and as shares of spending rather than as ratios to GDP (Table 5). As Section 4.1 makes clear, this costs us nothing for the war itself: with no marketable debt stock, the GDP-scaled terms of the decomposition vanish, and the financing shares we report require no GDP at all.

Three measurement choices shape what follows, and we flag them now. First, debt is valued at par before 2000 and at market thereafter, since market prices exist only once the bond market does; the implicit-interest channel of Section 2 is therefore measurable only in the modern period. Second, every series is spliced across sources at well-defined dates, and the 1953 and 1962 currency reforms are linked (Appendix A documents each join). Third, and most important, the accounting identity does not close exactly: the change in measured debt differs from the accumulated flow deficit by a stock–flow adjustment—the residual ε_t of equation (2) (Escolano, 2010). Part of it is economically real—foreign aid financed developmental deficits without raising domestic debt, and the 1997 guarantees sit off the books—and part is measurement, from the proxy for r^B and the splices. We do not paper over it: we report it as a residual and return to it with the decomposition (Section 3).

Figure 1 shows the fiscal setting in which these shocks occurred. Panel (a) plots central-government expenditure and revenue as shares of GDP: a small government in the 1950s—spending about a tenth of GDP, revenue less—grows into a modern fiscal state collecting and spending a quarter of GDP, with expenditure above revenue through most of the period. Panel (b) plots the debt this financed: central-government debt (concept D1, excluding guarantees), held near or below 10% of GDP through the development decades and rising to roughly half of GDP since. The monetary environment ran the other way and is taken up where it bears on financing—a wartime hyperinflation that paid for the Korean War (Section 4.1), high but declining inflation through the development decades that quietly eroded the debt in panel (b) (Section 3), and low, stable inflation since.

A second fact is equally central: Korea had no marketable long-term government bond market before about 2000. The earliest Korea Treasury Bond benchmark yields begin in 1995 (3- and 5-year), with 10-, 20-, 30-, and 50-year maturities added only in 2000, 2006, 2012, and 2016 respectively. The sovereign bond market is, in effect, a creation of the post-2000 period—which is precisely why the financing of earlier shocks could not rely on it.

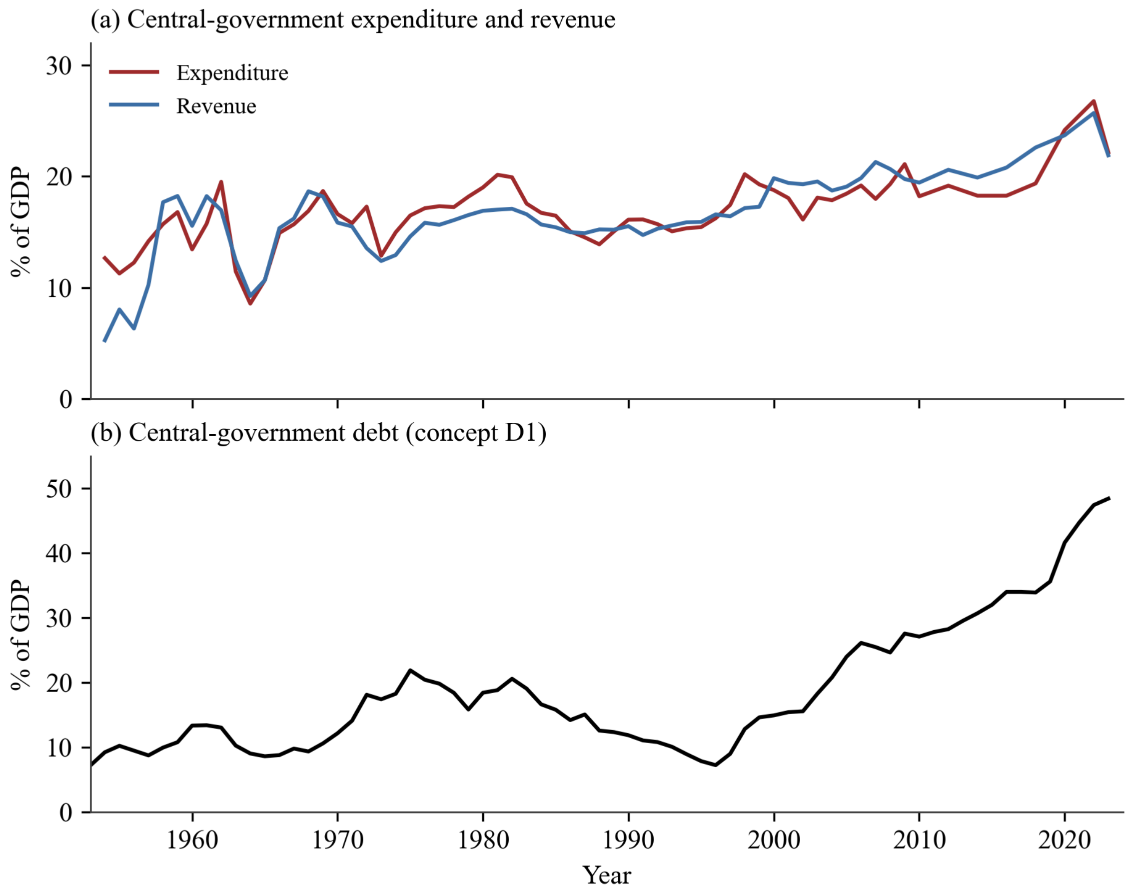

Figure 1: The fiscal setting, 1954–2023. (a) Central-government expenditure and revenue, % of GDP, splicing the Naksungdae Institute series (to 2011) to official modern accounts as a single continuous series; the Korean War years, for which no won GDP exists, are accounted separately in levels (Table 5). (b) Central-government debt (concept D1, which excludes government-guaranteed debt), % of GDP.

3 How Korea financed its government, 1961–2023

This section reads Korea's budget constraint as a continuous record. Rather than open with the crises, we open with the decades they punctuate, putting a single question to every year from 1961 to 2023: how did Korea pay for its government? We apply the law of motion (2), which splits the annual change in central-government debt/GDP (concept D1) into five forces—the primary deficit, interest, real growth, inflation, and seigniorage—plus a stock–flow residual. We read the result off two views of the same calculation: Figure 2 year by year, and Figure 4 averaged over five crisis-bounded periods.

Table 1: Data sources and coverage

Series	Source	Coverage
Wholesale / consumer prices	BOK ECOS	1910–2026
Nominal & real GDP	BOK ECOS / Naksungdae	1953–2023 (colonial 1911–40)
Central government debt	Naksungdae T27	1948–2023
of which marketable bonds (*kukch'ae*)	Naksungdae T27	1950–2023
Government revenue & expenditure	Naksungdae T25	1954–2023
Base money (M_0)	BOK 2005 / BOK ECOS	1950–2023
KTB yields, maturities 1–50yr	BOK ECOS	1995–2026
Bank lending rate / bond yield	BOK 2005 / BOK ECOS	1945–2023
War-era G, T, note issue	BOK 2005	1948–1960

Notes: "BOK 2005" is Bank of Korea, *Sixty Years of Liberation in Numbers* (2005) (Bank of Korea, 2005), the wartime fiscal–monetary compendium underlying Kim and Lee (2010). "Naksungdae" refers to the long-run *Historical Statistics of Korea* (Kim et al., 2018), spliced to official series (debt concept D1, the managed balance) from 2016. "ECOS" is the Bank of Korea's Economic Statistics System.

Both figures sign each force by whether it raises or lowers the debt ratio. In Figure 2, each year stacks the debt-raising contributions (primary deficit and interest) above zero and the debt-reducing ones (real growth, inflation, and seigniorage) below; the black line is the actual change in debt/GDP, equal to the net of the bars up to the stock–flow residual. Figure 4 collapses the same contributions into five crisis-bounded period averages, with the residual drawn explicitly so the bars sum exactly to the actual mean change (the black diamond). Table 3 reports those averages as numbers.

3.1 The whole period at a glance

Read across the whole period, the most arresting fact about Korea's public debt is how little it moved: averaged from 1961 to 2023, the debt ratio rose by barely half a point of GDP a year (Figure 3, and the final column of Table 3). Interest and a stock–flow residual together pushed it up by about 3.5 points a year; real growth (-1.5), inflation (-1.2), and seigniorage (-1.1) pulled it down by almost as much; the primary deficit was small ($+0.6$). This calm average, though, is the sum of a rise and a fall—of a single engine that held the debt down for three decades and then died.

3.2 The inflation-tax engine, 1961–96

Through the development decades the debt stood still, and not for want of borrowing (Figure 4). Deficits of about 0.9 points of GDP a year, carried at double-digit nominal interest ($+3.0$), would on their own have driven the ratio up by nearly four points a year; instead rapid real growth (-2.0), high inflation (-2.0), and seigniorage (-1.7) eroded it by 5.7, and the ratio ended the period in the mid-teens, almost exactly where it began (+0.1 net, against a positive stock–flow residual we return to below). An engine of growth and inflation, in short, ran fast enough to absorb everything borrowing threw at it.

In the decade before the Asian crisis the same engine, now joined by budget discipline, did more than hold the debt down—it pushed it down. The developmental inflation had subsided (its contribution falls to −0.3), but rapid growth (−1.8) and seigniorage (−0.9) still eroded the ratio, and with the budget in primary *surplus* (−0.1) debt fell by about 0.8 points a year. This is the one stretch in seventy years in which Korea's debt ratio declined.

What did the eroding, and on whom, is where Korea parts company with the textbook—and it is the central empirical finding of these decades. In the post-war United States and United Kingdom, a large stock of government debt was slowly liquidated by sustained negative real interest rates—financial repression in the sense of Reinhart and Sbrancia (2015). Korea did not work this way. Its real interest rate on debt was negative in only five years of the period—1963–64, 1974–75, and 1980, each an inflation spike tied to a devaluation or an oil shock. For the rest, formal rates ran above inflation: the 1965 interest-rate reform deliberately raised deposit and lending rates to mobilize saving (McKinnon, 1973; Shaw, 1973), the opposite of repression on rates. The erosion came instead from two sources that require no negative real rate—seigniorage, the inflation tax on base money, which averaged about 1.7 to 2.2% of GDP a year, and rapid real growth working through the denominator. Korea leaned on the inflation tax, but it fell on holders of money, not on bondholders—not because Korea had issued no bonds, but because its bond market was small and immature. The debt itself was dominated by foreign borrowing (around half) and domestic loans; marketable bonds were only a tenth to a fifth of it, and total debt only about 15% of GDP, so the traded bond stock was on the order of 3% of GDP—and those bonds were largely compulsory placements (Section 2.2) rather than a deep market whose real value inflation could erode. There was simply little bondholder wealth to tax.

Table 2 puts numbers on the contrast. Korea's debt was an order of magnitude smaller than the post-war US and UK stocks, real rates were negative in only a fifth of the years rather than half, and debt liquidation was negligible—against the 2 to 3 percent of GDP a year that negative real rates transferred from US and UK bondholders to their treasuries (Reinhart and Sbrancia, 2015). What Korea did levy fell on money: seigniorage of nearly 2 percent of GDP a year, not the slow liquidation of a large bond stock. One caveat keeps us honest—the pre-2000 bond rate is proxied by the lending rate, and the true coupon on Korea's repressed bonds was probably lower, so the table understates Korean debt liquidation; seigniorage, which needs no rate assumption, is unaffected.

Table 2: Debt liquidation versus seigniorage: Korea's development era against the post-war United States and United Kingdom

	Avg. debt/GDP (%)	Years with neg. real rate	Debt liquidation (% GDP/yr)	Seigniorage (% GDP/yr)
United States, 1945–80	≈50	≈50%	2.3	<0.5
United Kingdom, 1945–80	≈120	≈60%	3.0	<0.5
Korea, 1961–85	15	20%	≈0	1.9

US and UK: Reinhart and Sbrancia (2015). Korea: authors' calculation from the decomposition of Figure 2, with debt liquidation = $-$(interest + inflation) contributions and seigniorage = ΔM_0/GDP; the pre-2000 bond rate is proxied by the lending rate.

3.3 The engine dies, 1997–2023

Around 1997 the engine died, and the debt began a climb it has not since reversed. Growth halved (-0.9), and inflation and seigniorage fell to near zero (-0.3 and -0.2): the erosion that had offset deficits for thirty-five years was gone. Debt rose by 1.7 points of GDP a year. Yet the budget was still in primary *surplus* (-0.6); the rise came almost entirely from the stock–flow residual ($+2.7$)—the off-budget public funds of the 1997 crisis converted into national debt, together with the bonds Korea issued to build its foreign-exchange reserves (Section 4.2). The debt of one crisis surfaced on the books just as the inflation tax that might have eroded it disappeared.

After 2008 the new regime settled into a quiet, steady rise. With the inflation tax retired and the 1997 legacy largely absorbed, no single force dominated; debt drifted up by half a point a year ($+0.5$), the residual now smaller ($+1.4$) and reflecting continued reserve and financial-sector issuance.

COVID completed the inversion. For the first time the primary deficit itself—not the absence of the inflation tax—drove the debt: the primary balance swung to a deficit of nearly five points of GDP a year, and debt rose by 2.8 points a year, the fastest in the record (Section 4.4).

Read as a question of *incidence*, the decomposition tells a story of who has paid for the Korean state. In the development decades the burden fell quietly on holders of money, through a seigniorage tax that never appears as a line in the budget—and, in the war and the inflation spikes, on the captive holders of below-market bonds. In the modern period it falls visibly: on taxpayers, who fund the primary surpluses, and on the holders of marketable government debt. The same fiscal pressure, borne by different people as the instruments of finance changed. And because the engine is now gone, the aging-driven deficits ahead have no inflation tax to absorb them—the question of Section 5.

The decomposition does not close exactly, and the gap is worth confronting directly. A stock–flow residual averaging about two points of GDP a year separates the measured change in debt

Table 3: Average contributions to the annual change in debt/GDP, by period (pp of GDP per year)

	1961–85	1986–96	1997–2007	2008–19	2020–23	1961–2023
Primary deficit	0.9	−0.1	−0.6	0.4	4.9	0.6
Interest	3.0	1.6	1.0	0.8	0.9	1.9
Real growth	−2.0	−1.8	−0.9	−0.7	−0.1	−1.5
Inflation	−2.0	−0.3	−0.3	−0.8	−1.6	−1.2
Seigniorage	−1.7	−0.9	−0.2	−0.6	−0.9	−1.1
Residual (SFA)	2.0	0.7	2.7	1.4	−0.4	1.6
Net change	**0.1**	**−0.8**	**1.7**	**0.5**	**2.8**	**0.5**

Complete-case years (all five terms observed), 1961 onward. Columns may not sum exactly to the net change because of rounding. Source: decomposition of Figure 2.

from the identified flows—larger, in several periods, than the net change it helps to explain. Such residuals are not peculiar to Korea: across countries it is the stock–flow adjustment, rather than the primary deficit, that drives large debt movements (Campos et al., 2006; Andrian et al., 2025), and it reflects a recognized set of forces—valuation effects, the net acquisition of financial assets, and the realization of contingent liabilities (Weber, 2012; Bova et al., 2019). Korea's residual decomposes along exactly these lines (Table 4). Foreign-currency revaluation accounts for about half a point a year in the development decades, when half the debt was external; reserve accumulation—the bonds issued to purchase foreign exchange, a net acquisition of financial assets—accounts for some 1.5 points a year in 1997–2007 and 0.5 thereafter; and the 1997 crisis added the off-budget public funds (guaranteed KAMCO and KDIC bonds) that later surfaced as national debt, the contingent-liability realization that Bova et al. (2019) document for Korea among others. A remainder of about a point a year—developmental on-lending in the early decades, and measurement from the interest-rate proxy and the currency-reform splices—we leave explicitly unallocated, in the spirit of the "other means" line that Hall and Sargent (2022) themselves carry. The one period the decomposition cannot cover at all is the Korean War, for which no won GDP exists; it is handled in levels (Table 5, Section 4.1).

4 The shocks: the regime at its sharpest

Against this secular backdrop, the three great shocks were not exceptions to Korea's financing regime so much as its most concentrated expressions: each was paid for in the dominant mode of its era, only more so. The Korean War distilled the inflation tax of the early developmental state; the 1997 crisis, the off-budget contingent debt that standard accounts miss; COVID, the marketable bond finance of a mature Treasury market. We take them in turn. For the war and for

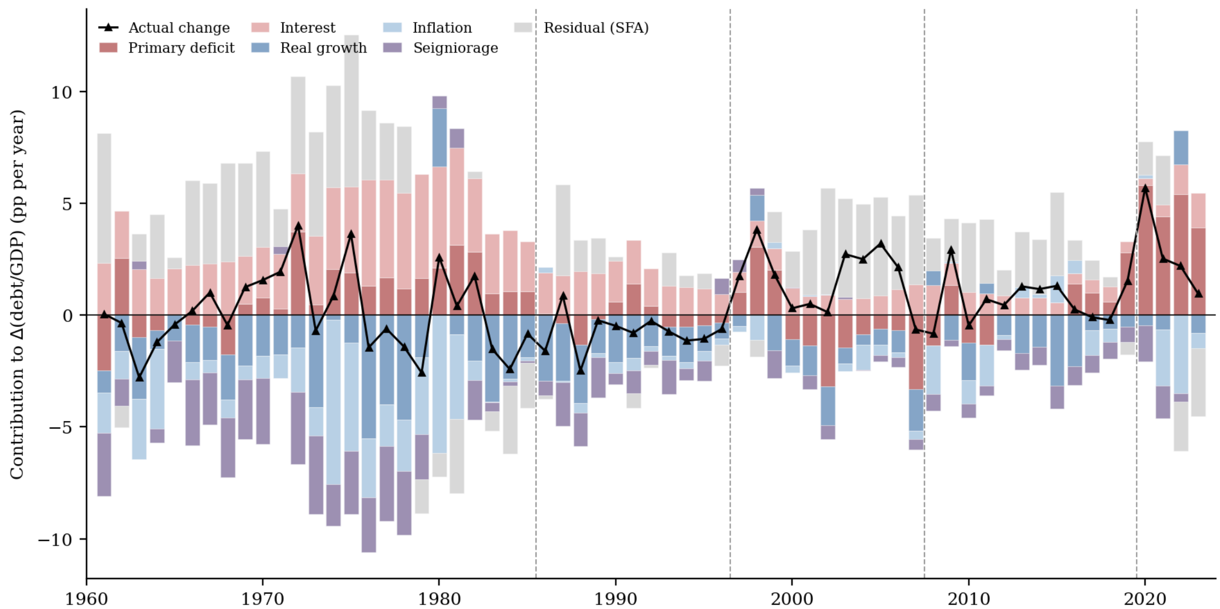

Figure 2: Annual change in Korea's central-government debt/GDP (concept D1, excluding guarantees), 1961–2023, decomposed by equation (2) into the primary deficit, nominal interest, real growth, inflation, and seigniorage, plus the stock–flow residual (grey). Bars are contributions (pp of GDP); the residual is drawn explicitly so the stacked bars sum exactly to the actual change (black line). Dashed vertical lines mark the five-period boundaries of Table 3. Two features stand out: the large negative inflation-and-seigniorage block (light blue, purple) that held the debt down through the development decades, and the large positive residual in those same decades and around 1997—the off-budget and foreign-currency financing discussed in the text—which shrinks to near zero in the clean modern period. The Korean War (1950–52) is omitted—no won-denominated GDP exists for those years, so it is accounted in levels (Table 5). Sources: Naksungdae Institute (debt, fiscal, to 2015) spliced to official D1 and the managed balance; Bank of Korea (money, prices, yields).

COVID, equation (3) splits the spending surge above a pre-shock baseline into the shares paid by taxes, marketable bonds, money creation, and—for the war—foreign aid; the war is the limiting case in which, with no bond market and no usable GDP, the surge is measured against spending itself rather than GDP (Table 5). The 1997 crisis is the exception that proves the rule: the same decomposition, applied to the standard budget, shows almost nothing because the financing was off the books, so we account for it directly through the public-funds ledger (Table 7) instead.

4.1 The Korean War, 1950–53: the inflation tax

The war was fought on an almost empty fiscal base. Korea did issue government bonds—the *Foundation Bonds* (*Kŏn'guk kukch'ae*), authorized in December 1949 and first sold in January 1950—but these were small, largely compulsory placements with no secondary market. Table 5 shows the result: of cumulative war spending, taxes covered 38% and marketable bonds just 4%. The rest was met by central-bank borrowing—35%, mostly Bank of Korea advances, which is money

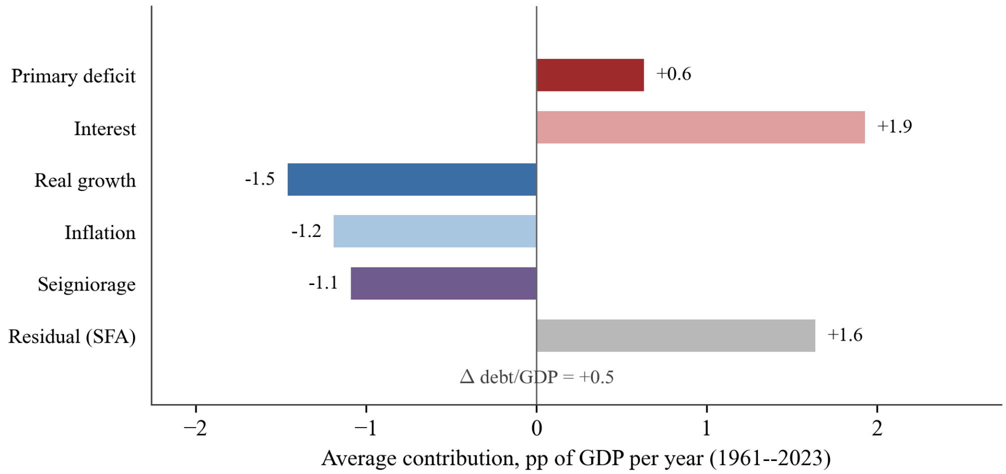

Figure 3: Average contribution of each force to the annual change in debt/GDP over the whole period, 1961–2023 (pp of GDP per year). Debt-raising forces extend right, debt-reducing forces left; their sum is the net change of $+0.5$ points a year. Bars are the row means of Table 3, final column.

creation—and by U.S. aid, 23%.

The monetary consequence was an explosion in prices. The wholesale price index rose about eighteenfold between the outbreak of war in June 1950 and the armistice of July 1953—roughly 155% a year, with year-on-year inflation peaking near 380% in mid-1951. With almost no marketable nominal debt to revalue, the inflationary finance worked through seigniorage—the inflation tax on money and on the monetized central-bank advances—not through the revaluation of traded bonds.

The compulsory Foundation Bonds are, in Hall and Sargent's vocabulary, *financial repression*: a captive class of subscribers—banks, importers, firms—was made to hold low-coupon claims at below-market terms during a hyperinflation, bearing large negative real returns, a discriminatory tax on a specific creditor class rather than a voluntary loan (Hall and Sargent, 2014). The contrast with the United States is sharp. Hall and Sargent find the U.S. financed its own involvement in the Korean War almost entirely through taxes (Hall and Sargent, 2021); Korea, lacking the fiscal and financial capacity to do likewise, paid through the inflation tax and foreign aid.

4.2 The 1997 crisis: off the books

Applying (3) to standard central-government accounts "fails": the measured spending surge is small (about 3% of GDP) and the decomposition leaves a large residual—which is exactly the point. Korea resolved the crisis not mainly through the central-government budget but through guaranteed bonds issued by the Korea Asset Management Corporation (KAMCO) and the Korea Deposit Insurance Corporation (KDIC). These "public funds" totaled **168.5 trillion won** injected

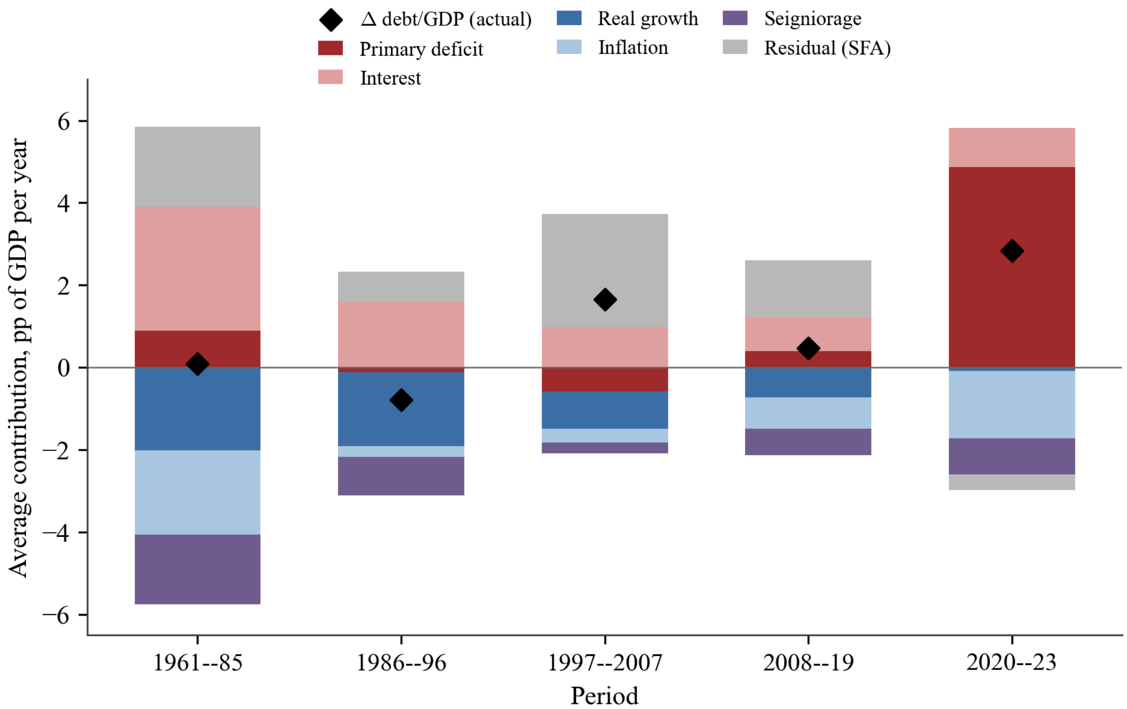

Figure 4: The same decomposition averaged over five crisis-bounded periods (pp of GDP per year). Bars are the mean contributions of equation (2); the black diamond is the actual mean change in debt/GDP, to which the bars (including the stock–flow residual) sum. The development era (1961–85) sets a large debt-raising block against an even larger eroding one, netting to zero; debt then *falls* under the budget discipline of 1986–96; the 1997–2007 rise is driven by the stock–flow residual—the off-budget public funds converted to national debt, and reserve accumulation; and COVID (2020–23) is the first period in which the primary deficit itself dominates. Complete-case years only, 1961 onward.

from late 1997 (Table 7), on the order of **30% of a year's GDP**—an order of magnitude larger than the on-budget surge, and almost entirely invisible in headline central-government statistics. Because it is off the books, almost none of it enters the standard decomposition of Figure 2—which is exactly why that decomposition, on its own, understates the crisis. Korea financed its financial crisis through contingent, state-guaranteed debt—a mode of financing distinct from both the inflation tax of the Korean War and the marketable bonds of COVID.

Table 6 makes the gap concrete by running equation (3) on the on-budget accounts, exactly as for the war and COVID. The total on-budget cost of 1997–98 is only about 6% of GDP—met chiefly by borrowing and dominated by the recession itself, output fell so sharply that the growth term *added* to the burden—with taxes covering less than a third and inflation a minor role. Against this 6% sits the 168.5 trillion won of off-budget public funds, roughly 30% of GDP: five times larger, and invisible to the decomposition. The accounting that works for the war and for COVID barely registers the 1997 crisis; to see it, one must look off the budget, where Table 7 looks.

Table 4: Decomposing the stock–flow residual, by period (pp of GDP per year)

	Residual (SFA)	FX valuation	Reserve accumulation	Other
1961–85	1.9	0.5	—	1.4
1986–96	0.7	−0.1	0.1	0.7
1997–2007	2.8	0.0	1.5	1.3
2008–19	1.7	0.0	0.5	1.2
2020–23	−0.4	—	1.0	−1.4
1961–2023	1.7	0.2	0.9	0.6

"FX valuation" is the won revaluation of foreign-currency debt (won depreciation × the foreign-currency share of debt); "reserve accumulation" is the annual change in foreign-exchange reserves (a net acquisition of financial assets financed by issuance); "other" collects the 1997 off-budget public-funds conversion (a contingent-liability realization, Bova et al., 2019), developmental on-lending, and measurement (the interest-rate proxy and the currency-reform splices). Sources: ECOS (reserves, exchange rate); decomposition of Figure 2.

Table 5: Financing of the Korean War, 1950–53 (cumulative; billion won and as a share of total war spending, $\sum G = 100$). Shares are of spending, *not* of GDP as in the COVID accounting of Table 9, because won-denominated GDP does not exist for 1950–52 (Section 2.2); these shares require no GDP. Sources: Bank of Korea 2005, Naksungdae Institute.

	Amount	% of spending
Government spending ($\sum G$)	9.08	100
Taxes ($\sum T$)	3.46	38
Marketable bonds (Δ *kukch'ae*)	0.34	4
Central-bank borrowing / money creation (Δ *ch'aibgŭm*)	3.17	35
Aid / other (residual)	2.11	23
Memo: total debt rise ΔB	3.52	39
Memo: seigniorage—the inflation tax (ΔM_0)	2.20	24
Memo: wholesale prices, Jun 1950 → Jul 1953		×18 (≈155%/yr)

The right accounting object here is the one Hall and Sargent use for credit that is "backed" by an asset of uncertain value: net debt $B - C$, where B is the guaranteed bonds issued and C is the present value of recoveries on the underlying impaired assets. The *ex post* fiscal cost is then B minus what was actually recovered. For Korea's public funds, recoveries reached 120.4 trillion won by 2023, so the realized cost was about $168.5 - 120.4 \approx 48$ trillion won—roughly 2% of current GDP, or near 10% of 1998 GDP. The headline 30%-of-GDP injection was thus a gross flow largely revolved back; the true subsidy was the un-recovered remainder. Either way, none of it appears in the standard central-government deficit of 1997–98.

Table 6: The 1997 crisis through equation (3): on-budget financing, 1997–98 (deviation from the 1992–96 baseline). The standard decomposition registers a total cost of only about 6% of GDP—dominated by the recession (the large negative growth contribution) and met chiefly by borrowing—a small fraction of the true crisis cost, which was carried off the budget (Table 7).

	pp of GDP	% of cost
Spending surge + interest (total cost)	6.0	100
Taxes	1.9	31
Borrowing	4.4	73
Money / seigniorage	2.1	34
Inflation dilution	0.8	14
Growth dilution / residual	−3.1	−52
Memo: off-budget public funds	168.5 tn won ≈ 30% of GDP	

Table 7: Public funds: off-budget financing of the 1997 crisis (trillion won; Public Fund Oversight Committee)

	Amount
Equity injection (*ch'ulcha*)	110.7
Asset purchase (*chasan maeip*)	38.6
Other (contributions, deposit payoff)	19.1
Total injected	**168.5**
≈ % of a year's GDP	≈30
recovered by 2023	120.4 (71%)

4.3 The 2008–09 global financial crisis: bond finance arrives

By 2008 the transition was complete, and the global financial crisis is the proof. Korea met its stimulus—a spending surge of about 4% of GDP—almost entirely with marketable borrowing (Table 8): bonds covered nearly all of it, taxes about a quarter, and money creation none. The deep Treasury market that was absent in 1950 and improvised around in 1997 now simply absorbed the shock; the inflation tax played no part. The episode also shows the interest-rate-risk channel of Section 2 running in the government's *favour*: as the Bank of Korea cut rates and investors fled to safety, bond yields fell, the price of outstanding debt rose, and bondholders earned a real holding return near +14% in 2008, raising the government's real cost of funds. COVID, as the next subsection shows, would run the same conventional financing through the opposite rate move.

4.4 COVID-19, 2020–22: conventional bond finance

With a deep Treasury market in place, COVID was financed conventionally. Table 9 decomposes the surge: of the spending increase over 2020–22 relative to the 2015–19 baseline, taxes financed

Table 8: Financing of the 2008–09 global financial crisis (deviation from the 2004–07 baseline, % of GDP). Like COVID and unlike the earlier shocks, the surge is met almost entirely by marketable borrowing.

	pp of GDP	% of cost
Spending surge + interest (total cost)	3.9	100
Taxes	0.9	24
Borrowing	3.8	98
Money / seigniorage	−0.4	−10
Inflation dilution	1.1	27
Growth dilution / residual	−1.5	−39

about 49% and marketable borrowing about 45%, with money creation (6%) and inflation a minor role. Notably, Korean revenue *rose* relative to GDP during the pandemic (from 26% to 30%), the opposite of the U.S. experience.

Table 9: Financing of COVID-19, 2020–22 (deviation from 2015–19 baseline, % of GDP)

	pp of GDP	% of cost
Spending surge + interest (total cost)	18.41	100
Taxes	8.98	49
Borrowing	8.19	45
Money / seigniorage	1.08	6
Inflation dilution	2.63	14
Growth dilution / residual	−2.47	−13

Unlike the Korean War, COVID activated the second inflationary channel: *debt revaluation*. As inflation rose to 5% and the 10-year KTB yield climbed from 1.5% (2020) to 3.6% (2023), holders of long Korean government bonds earned sharply negative real returns—about −13% in 2022 alone and roughly −17% cumulatively over 2020–23 on a constant-maturity 10-year position. The state thus imposed part of the COVID cost on its bondholders—exactly the capital levy that was impossible in 1950 for want of a bond market, and the mirror image of 2008–09, when the same channel had handed those bondholders a windfall instead.

Who paid? The composition of bondholders gives the levy a distinctly Korean incidence. Insurance companies and pension funds hold the largest share of Korea Treasury Bonds (on the order of one-third, the single biggest being the National Pension Service), with banks (∼17%) and foreign investors (∼18%) next. The 2022 real loss therefore fell substantially on the National Pension Service—making it, in part, an *intergenerational* transfer from future pensioners—and on foreign holders, a channel with no analog in the closed wartime capital market.

Looking forward, the COVID expenditure surge has partly persisted while revenue has not, so the consolidated budget constraint implies future adjustment. Here Korea differs institutionally from both its own past and the U.S. case of Hall and Sargent (2025): an inflation-targeting central bank has all but foreclosed the seigniorage and debt-revaluation margins on which the developmental state once leaned. With the inflation tax off the table and an aging population pushing projected debt/GDP steadily upward, the arithmetic must close through taxes or spending. The same fiscal-capacity development that gave Korea a bond market also took away its historical escape valve.

4.5 Taking stock: the shifting financing mix

Figure 5 places the episodes side by side and adds two U.S. benchmarks. Korea's financing of fiscal shocks moves from monetized borrowing and aid in the Korean War (marketable bonds just 4%), through off-budget contingent debt in 1997 (not shown in the bar because it is off the books), to a near-even split of taxes and marketable bonds in COVID. The United States, by contrast, relied on marketable debt in both World War II and COVID, and—as Hall and Sargent show—repaid bondholders partly through subsequent inflation. The same kind of shock is financed by different means, and the means track the development of the state's fiscal and financial capacity: a marketable government bond market simply did not exist in Korea until it was built.

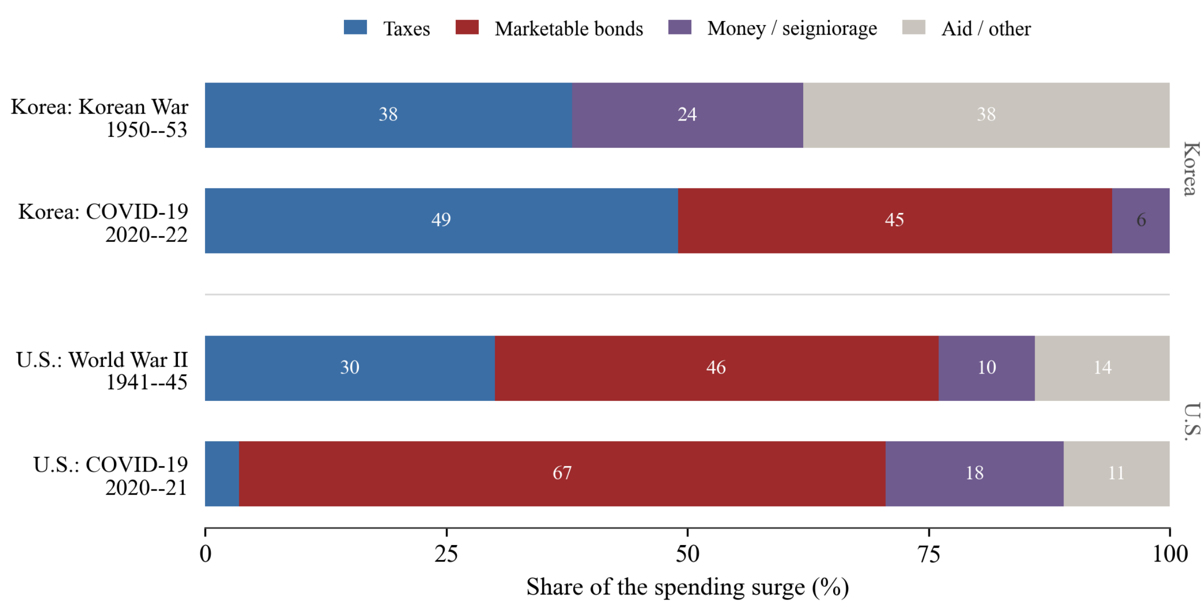

Figure 5: Financing of fiscal shocks as a share of the spending surge, Korea versus the United States. Korea shifts from inflation- and aid-financed war (1950–53; no marketable bonds) to conventional tax-and-bond finance under COVID; the United States relies on marketable bonds throughout. The 1997 crisis, financed off the budget through guaranteed debt (Section 4.2), does not fit the spending-surge frame and is omitted here. Korea: Tables 5 and 9. United States: Hall and Sargent (2021, 2022).

5 The future: the erosion engine switched off

The same accounting that explains the past can be pointed at the future. We do so with the National Assembly Budget Office's seventh long-term fiscal projection (NABO, 2025), an official, legislatively mandated forecast that carries Korea's population, economy, and budget to 2072 under a demographic baseline (Statistics Korea's medium-variant population) and a no-policy-change fiscal rule. We use its central, population-median scenario, and report its low- and high-population variants as a sensitivity band.

The projection supplies exactly the inputs the law of motion (2) requires, which is why we can apply the identical accounting to the future. From NABO we take the projected annual paths of six series: real GDP growth, the GDP deflator (our inflation π), nominal GDP, the three-year Treasury bond rate (our proxy for the average return r^B on debt), the managed fiscal balance (our primary deficit $G - T$), and debt/GDP itself, which serves as a check on the accumulated flows. The one term NABO does not project, seigniorage, we compute as before, from base-money growth at the projected nominal rate with the base-to-GDP ratio held at its recent level. Table 10 collects the key assumptions; their defining feature is the disappearance of the forces that once held the debt down—growth slows to a crawl, inflation is anchored at target, and the Treasury rate sits right at the now-low nominal growth rate. One simplification we flag: with no maturity structure projected, the interest term uses the three-year rate as the average debt return, so the projection omits the bond-revaluation channel that mattered in 2022 (Section 4.4)—a reasonable omission, since NABO assumes no inflation surprises.

Table 10: Key assumptions of the NABO long-term projection (population-median scenario)

	2025	2072	Avg. 2025–72
Real GDP growth (%)	2.2	0.3	1.0
Inflation, GDP deflator (%)	2.0	2.0	1.9
Nominal GDP growth (%)	4.2	2.1	2.9
Three-year Treasury bond rate (%)	2.5	2.1	2.2
Growth-adjusted rate $i - g^{\text{nom}}$ (pp)	-1.7	0.0	-0.7
Mandatory (aging-driven) outlays (% GDP)	14	22	—
Managed fiscal balance (% GDP)	-3.2	-6.4	—
Debt/GDP (%)	47.8	173.0	—

Source: NABO (2025), population-median scenario. Under the low- and high-population variants, 2072 debt/GDP instead reaches 182% and 163%. Seigniorage is not a NABO input; we compute it from projected base-money growth (Section 2.2). The growth-adjusted rate $i - g^{\text{nom}}$ is the three-year Treasury rate less nominal GDP growth.

The result is the mirror image of the development decades (Figure 6). Debt/GDP more than triples, from 48% in 2025 to 173% in 2072, and the decomposition shows why. The rise is driven almost entirely by the primary deficit—the structural gap that opens as an aging population lifts mandatory spending from 14% to 22% of GDP. The forces that held debt down in the 1960s and 1970s are gone: with inflation at 2% rather than 15–40%, and the Treasury rate sitting right at the now-low nominal growth rate, the growth-adjusted interest rate $i - g$ hovers near zero instead of the deeply negative values of the development era (Table 11). Seigniorage, which delivered nearly 2% of GDP a year then, delivers about 0.2% now. (Under NABO's low- and high-population scenarios, 2072 debt instead reaches 182% and 163% of GDP; a stock–flow residual of about 0.6 points a year, of the same order as in the historical accounting, reflects financing beyond the managed balance, chiefly the post-depletion social-security shortfall.)

This is the forward face of the paper's central finding. Korea financed its developmental state, and its first great shock, through an inflation tax that fell on holders of money. That tax is gone—retired by the very inflation-targeting regime and deep bond market that mark a mature fiscal state. The arithmetic that once let large deficits coexist with stable debt no longer holds: the aging-driven deficits ahead accumulate, point for point, into debt, and can be closed only through taxes and spending. The capacity that gave Korea a bond market took away its escape valve.

Table 11: Projected decomposition of the change in debt/GDP, 2025–2072 (pp of GDP per year), versus the development era. Source: equation (2) applied to NABO (2025); development-era row from Figure 4.

	Primary def.	Interest	Growth	Inflation	Seign.	$i - g^{\text{nom}}$
2025–40	2.4	1.3	−1.0	−1.1	−0.3	−1.5
2040–55	2.9	2.1	−0.9	−1.8	−0.2	−0.7
2055–72	2.8	3.0	−0.6	−2.5	−0.2	−0.2
cf. 1961–85	0.9	3.0	−2.0	−2.0	−1.7	≈ -7

6 Conclusion

Reconstructing Korea's government budget constraint across three fiscal shocks reveals a systematic migration in how the state pays for crises—from the inflation tax and forced bonds of the Korean War, through the off-budget contingent debt of 1997, to the conventional taxes and marketable bonds of COVID. A government can inflate away only the debt it has issued; as Korea built fiscal capacity and a sovereign bond market, the financing of shocks moved accordingly, and so did the incidence of their cost—from money-holders and captive subscribers under wartime financial repression, to bondholders (notably the National Pension Service and foreign investors) under COVID. The progression distinguishes a developing-country fiscal history from the bond-financed pattern of the United States, and it carries a forward-looking implication: the very capac-

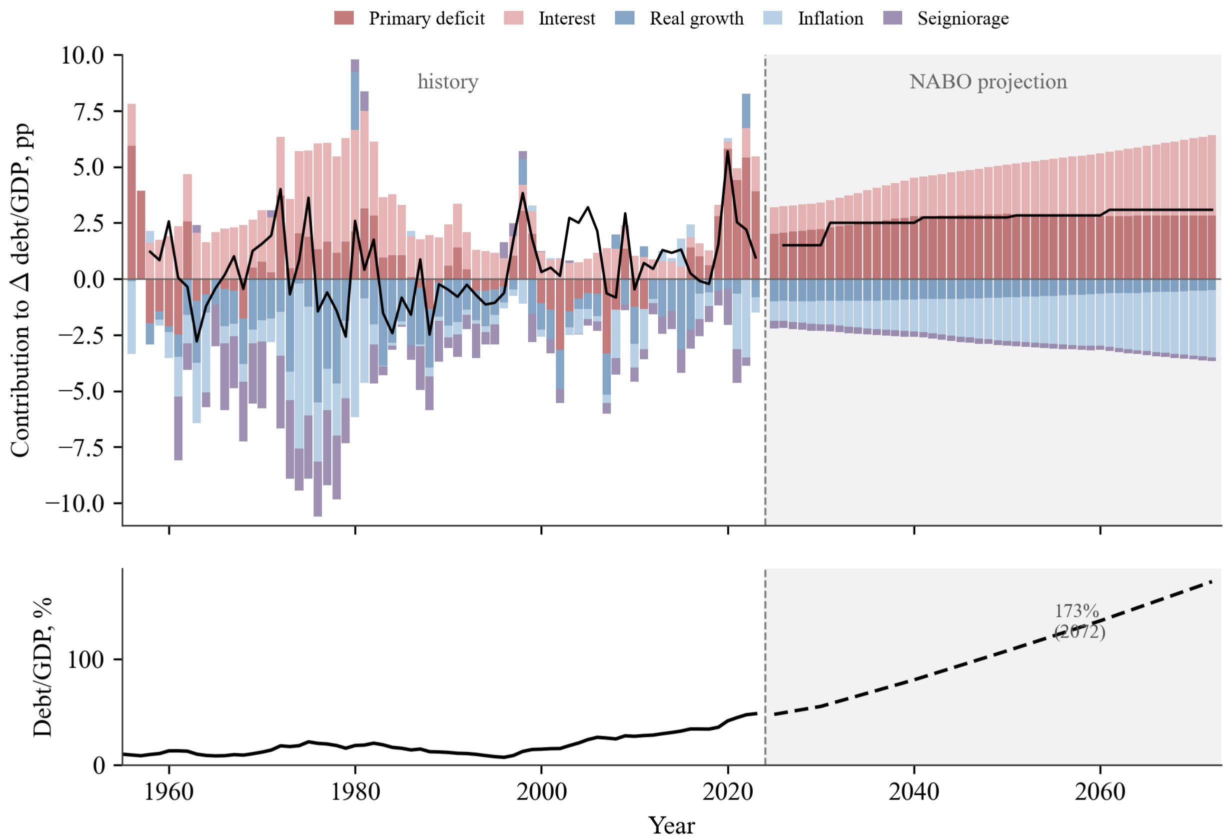

Figure 6: *Top:* annual change in central-government debt/GDP decomposed by equation (2), 1956–2023 (history) and 2025–2072 (NABO projection, shaded). In the development decades a large inflation-and-seigniorage block (light blue, purple) holds the actual change (black line) near zero; in the projection the inflation tax is gone and the primary deficit (dark red) drives debt steadily upward. *Bottom:* debt/GDP, rising to 173% by 2072. Projection: NABO (2025), population-median scenario; seigniorage from base-money growth at the projected nominal rate.

ity that gave Korea a bond market also retired the inflation tax, so its next fiscal adjustment must run through taxes and spending rather than the printing press.

A Data construction and sources

This appendix documents how each series entering the budget constraint is built, spliced, and valued, so that every figure in the paper can be traced to a primary source. The organizing principle is a single, consistent central-government concept, consolidated with the Bank of Korea, carried without conceptual break from 1948 to 2023. No one source spans that reach—it contains a war, two currency reforms, a developmental transformation, and the late birth of a bond market—so we splice across sources at well-defined dates, and we prefer Korean primary reconstructions to internationally harmonized databases, both because they reach further back and because they preserve the institutional detail (guarantees, foreign borrowing, note issue) on which our argument turns. The remainder of this section takes the series in turn.

Debt. The long-run statistics of the Naksungdae Institute (Kim et al., 2018) reconstruct central-government debt annually from 1948 and decompose it into marketable bonds (*kukch'ae*), domestic loans, foreign borrowing, and government-guaranteed liabilities. Our measure B is debt net of guarantees (concept D1). From 2016 we splice to the official D1 series of the Ministry of Economy and Finance; the two agree to within one or two points of GDP over their post-1990 overlap, the residual gap being local-government debt, which D1 excludes. We deliberately retain government-*guaranteed* debt as a separate, contingent series rather than folding it into B: it is the off-budget channel through which the 1997 crisis was financed (Section 4.2), reaching some 11% of GDP around 2000, and treating it as a contingent claim—debt only when called—rather than as ordinary borrowing is essential to measuring how that crisis was actually paid for.

Revenue and expenditure. The flows T and G come from the same Naksungdae source on a central-government basis, annual from 1954, with the war years from Bank of Korea (2005). The series is reported both gross and *net of foreign aid*, a distinction that lets us isolate the external term A_t of equation (1)—decisive for the Korean War, when grants covered close to a quarter of spending. Throughout, G is outlays net of interest and T is revenue net of transfers, so that the primary balance $G - T$ is the object that enters the decomposition; interest is handled separately through the return r^B of Section 2. From 2016 we use the managed-balance and consolidated central-government accounts of the Ministry of Economy and Finance, which align with the Naksungdae concept to within rounding over their overlap.

Money and prices. The monetary base M_0—the non-interest-bearing liability of the consolidated government, whose change is seigniorage—is continuous from 1960 (Bank of Korea, ECOS) and is extended through the war by the currency-issue figures in Bank of Korea (2005). For prices we use the Bank of Korea producer price index, which runs on a single, consistent concept from 1910 to the present (the historical series 404Y017 chained to its modern continuation) and so crosses the Korean War without a definitional break; its monthly version captures the wartime hyperinflation directly. Inflation π in the decomposition is the producer-price change; the consumer price index, available from 1965, gives very similar results in the modern period and serves as a cross-check.

Yields. Korea Treasury Bond benchmark yields are from ECOS: 3- and 5-year maturities from 1995, with 10-, 20-, 30-, and 50-year maturities added in 2000, 2006, 2012, and 2016 respectively—a span that is itself a record of the bond market's construction. For 1945 to the late 1990s, where no market yield exists, we use the Bank of Korea lending rate; it is an administered rate, but it is the relevant cost of funds in a period when government borrowing was largely from the central bank and the banking system. From 1987 the Naksungdae housing-bond yield provides an independent cross-check on the proxy, and the two move together.

Exchange rate and national accounts. These come from ECOS, with war-era figures from Bank of Korea (2005), the fiscal–monetary compendium underlying Kim and Lee (2010). One gap is irreducible: won-denominated GDP does not exist for 1941–52, because the war and the colonial-to-independence transition destroyed the statistical basis, and even the most complete long-run reconstruction (Kim, 2012) leaves the period blank, splicing the colonial accounts (to 1940) to the post-war Bank of Korea series (from 1953). This is why the continuous decomposition begins only in the mid-1950s and why the Korean War is accounted in levels (Appendix B).

Currency reforms and valuation. Two redenominations are linked across the sample: the 1953 reform replaced the won with the hwan at 100 : 1, and the 1962 reform replaced the hwan with the present won at 10 : 1. All nominal series are chained across both joins, and because the decomposition operates on ratios and growth rates the redenominations affect no result. Finally, debt is valued at par before 2000 and at market thereafter, since market prices exist only once a bond market does; the implicit capital-gain/loss channel is therefore measurable only in the modern period, a limitation we carry explicitly through the paper.

B The Korean War in levels

The continuous decomposition cannot cover 1950–53, because won-denominated GDP does not exist for those years (Kim, 2012). We therefore account the war in levels and as shares of spending, neither of which requires a GDP denominator. The treatment loses nothing of substance: with essentially no marketable debt stock, the GDP-scaled revaluation terms of equation (2)—growth dilution, inflation dilution, and the return on debt—vanish identically, so the only financing margins left are taxes, note issue, and foreign resources, and these we can measure directly in current hwan. The war is thus not a hole in the record but a limiting case of the same accounting, in which one financing mode does all the work.

The external resources of the war require the same care here as the contingent guarantees of 1997 do in the body. Following the foreign-credit accounting of Hall and Sargent (2019), we distinguish grants, which enter the constraint as a non-tax source of funds (A_t), from repayable U.S. and UN advances, which are an external liability backed by later won–dollar settlement; the economically relevant object is the liability net of what its backing ultimately delivers, so that the gross inflow of foreign resources during the war is not mistaken for its eventual fiscal cost. This is the same netting principle—an object of the form $B - C$—that we apply to the KAMCO and KDIC guarantees of 1997, and applying it consistently keeps the war's aid commensurable with the later crises' off-budget debt.

Using Bank of Korea (2005), we measure wartime government spending G, taxes T, foreign grants and advances A, and note issue ΔM in current hwan, and report each financing source as a share of spending (Table 5). Wholesale prices rose roughly eighteenfold between 1950 and 1953, so the inflation tax was extracted almost entirely from holders of money. The shares show a war paid for predominantly by note issue and foreign aid, with taxes covering only a small remainder—the inflation-tax mode in its purest form, levied by a state that had no bond market to borrow from, and the natural starting point of the migration the rest of the paper traces.

C The shock-financing decomposition

For each episode we apply the financing decomposition of equation (3), which sums deviations from a pre-shock "peacetime" baseline over the shock window and so isolates the marginal financing of the surge from the ordinary financing of the budget. Operationally, the baseline is the average of each financing flow over the years immediately preceding the window, and the windows we use are 1950–53 for the war, 1997–99 for the Asian crisis, 2008–09 for the global financial crisis, and 2020–22 for COVID-19. Within each window, equation (3) allocates the cumulative spending surge across explicit taxes, new marketable borrowing, money creation, and the erosion of outstanding debt by growth and inflation, with a cross term collecting the products of simultaneous deviations.

Three choices shape the result, and we make each explicit. The baseline length trades robustness against contamination: too short a window lets a single noisy pre-crisis year set the benchmark, too long a window lets secular trends masquerade as the baseline; a baseline of a few pre-shock years leaves the ranking of financing sources unchanged even as it shifts their levels slightly. The window end is set where the spending surge returns toward its baseline, not by a fixed horizon, so that a long crisis (1997, whose fiscal consequences ran for years) and a short one (the war) are each measured over their own duration. The cross term is small in every episode, and we report it separately rather than apportion it, so that the headline shares are not flattered by a discretionary allocation.

A virtue of the decomposition is that it is, by construction, invariant to the level of debt outstanding in years when the bond stock is negligible. In the war, where marketable debt is essentially zero, the revaluation terms drop out and the shares reduce to the division of spending among taxes, note issue, and aid—quantities that need no national accounts. The war's financing shares are therefore directly comparable to those of the modern episodes despite the missing GDP, and it is this comparability that lets Section 4 read all three shocks on a single scale.

D Robustness

Four measurement choices could in principle move the results, and we record here why none overturns the paper's conclusions. The first two concern how debt is valued and defined. Valuing pre-2000 debt at par rather than at market cannot bias the development-era findings, because no market existed to provide an alternative and the bond stock was in any case tiny; the choice begins to matter only from 2000, where market prices are used throughout and the implicit-return channel becomes measurable. And while our baseline is central-government debt (concept D1), the broader general-government measure D2—which adds the social-security funds, chiefly the National Pension Service—is reported as a supplement; the two measures diverge materially only after 2000 and leave the development-era and crisis decompositions unchanged, so the choice of concept does not drive the migration we document.

The remaining two choices concern prices and the residual. We use the producer price index for its continuity back to 1910, which is what lets a single inflation concept span the war; the consumer price index, available only from 1965, yields very similar inflation contributions over the modern period, so the developmental erosion we attribute to inflation is not an artifact of the index. Finally, we report the stock–flow residual ε_t rather than allocate it, precisely because much of it is economically meaningful rather than noise: its large positive values over 1997–2007 are the off-budget public funds converted to national debt and the bonds issued to build foreign-exchange reserves (Section 4.2), and burying these in an adjusted series would hide exactly the off-budget financing the paper sets out to measure. Reporting the residual openly is thus not a confession of imprecision but part of the accounting.

References

Bank of Korea (2005). *Sixty Years of Liberation in Numbers*. Seoul: Bank of Korea.

Escolano, J. (2010). "A Practical Guide to Public Debt Dynamics, Fiscal Sustainability, and Cyclical Adjustment of Budget Balances." IMF Technical Notes and Manuals 2010/02.

Hall, G. J., and T. J. Sargent (1997). "Accounting for the Federal Government's Cost of Funds." *Economic Perspectives*, Federal Reserve Bank of Chicago.

Hall, G. J., and T. J. Sargent (2014). "Fiscal Discriminations in Three Wars." *Journal of Monetary Economics* 61, 148–166.

Hall, G. J., and T. J. Sargent (2019). "Complications for the United States from International Credits, 1913–1940." In *Debt and Entanglements Between the Wars*. IMF.

Reinhart, C. M., and M. B. Sbrancia (2015). "The Liquidation of Government Debt." *Economic Policy* 30(82), 291–333.

Sargent, T. J., and N. Wallace (1981). "Some Unpleasant Monetarist Arithmetic." *Federal Reserve Bank of Minneapolis Quarterly Review* 5(3).

Hall, G. J., and T. J. Sargent (2011). "Interest Rate Risk and Other Determinants of Post-WWII U.S. Government Debt/GDP Dynamics." *American Economic Journal: Macroeconomics* 3(3), 192–214.

Hall, G. J., and T. J. Sargent (2021). "Debt and Taxes in Eight U.S. Wars and Two Insurrections." In *The Handbook of Historical Economics*.

Hall, G. J., and T. J. Sargent (2022). "Three World Wars: Fiscal-Monetary Consequences." *PNAS* 119(18).

Hall, G. J., and T. J. Sargent (2025). "Fiscal Consequences of the US War on COVID." *International Economic Review*.

Kim, N. (2012). *National Accounts of Korea, 1911–2010*. Seoul: Seoul National University Press.

Kim, N., K. Park, I. Park, and M. Cha, eds. (2018). *Historical Statistics of Korea*. Naksungdae Institute of Economic Research. Data: `naksung.re.kr/statistics`.

Kim, K., and J. Lee (2010). "Monetary Policy of the Bank of Korea during the First 60 Years." *Seoul Journal of Economics* 23(4).

National Assembly Budget Office (2025). *2025–2072 NABO Long-term Fiscal Projection*. Seoul: National Assembly Budget Office.

McKinnon, R. I. (1973). *Money and Capital in Economic Development*. Washington, DC: Brookings Institution.

Shaw, E. S. (1973). *Financial Deepening in Economic Development*. New York: Oxford University Press.

Hall, G. J., and T. J. Sargent (2023). "Financing Big US Federal Expenditure Surges: COVID-19 and Earlier US Wars." Book chapter / working paper.

Besley, T., and T. Persson (2009). "The Origins of State Capacity: Property Rights, Taxation, and Politics." *American Economic Review* 99(4), 1218–1244.

Besley, T., and T. Persson (2011). *Pillars of Prosperity: The Political Economics of Development Clusters*. Princeton University Press.

Dincecco, M. (2011). *Political Transformations and Public Finances: Europe, 1650–1913*. Cambridge University Press.

Eichengreen, B., and R. Hausmann (1999). "Exchange Rates and Financial Fragility." In *New Challenges for Monetary Policy*, Federal Reserve Bank of Kansas City.

Eichengreen, B., R. Hausmann, and U. Panizza (2005). "The Pain of Original Sin." In *Other People's Money*, University of Chicago Press.

Reinhart, C. M., J. F. Kirkegaard, and M. B. Sbrancia (2011). "Financial Repression Redux." *Finance & Development* 48(2).

Cagan, P. (1956). "The Monetary Dynamics of Hyperinflation." In *Studies in the Quantity Theory of Money*, University of Chicago Press.

Click, R. W. (1998). "Seigniorage in a Cross-Section of Countries." *Journal of Money, Credit and Banking* 30(2), 154–171.

Polackova, H. (1998). "Contingent Government Liabilities: A Hidden Risk for Fiscal Stability." World Bank Policy Research Working Paper 1989.

Abbas, S. A., N. Belhocine, A. El-Ganainy, and M. Horton (2011). "Historical Patterns and Dynamics of Public Debt." *IMF Economic Review* 59(4), 717–742.

Chien, Y., H. Lee, and J. Lee (2025). "A Consolidated View of Public Finance in Korea." Working paper, Bank of Korea / Federal Reserve Bank of St. Louis.

Koh, Y. (2005). "Identifying and Dealing with Fiscal Risks in Korea." Korea Development Institute. *[year/venue to confirm]*

Bordo, M. D., and M. D. Levy (2021). "Do Enlarged Fiscal Deficits Cause Inflation? The Historical Record." *Economic Affairs* 41(1), 59–83.

Lucas, R. E., and N. L. Stokey (1983). "Optimal Fiscal and Monetary Policy in an Economy without Capital." *Journal of Monetary Economics* 12(1), 55–93.

Andrian, L. G., C. M. Rodríguez, and O. Valencia (2025). "Surges in the Shadows: Stock-Flow Adjustments and Public Debt Spikes." *Emerging Markets Review* 67. Inter-American Development Bank Working Paper 13725 (2024).

Campos, C. F. S., D. Jaimovich, and U. Panizza (2006). "The Unexplained Part of Public Debt." *Emerging Markets Review* 7(3), 228–243.

Weber, A. (2012). "Stock-Flow Adjustments and Fiscal Transparency: A Cross-Country Comparison." IMF Working Paper 12/39.

Bova, E., M. Ruiz-Arranz, F. Toscani, and H. E. Ture (2019). "The Impact of Contingent Liability Realizations on Public Finances." *International Tax and Public Finance* 26(2), 381–417.

Hilscher, J., A. Raviv, and R. Reis (2022). "Inflating Away the Public Debt? An Empirical Assessment." *Review of Financial Studies* 35(3), 1553–1595.

Reis, R. (2021). "The Constraint on Public Debt when $r < g$ but $g < m$." Working paper, London School of Economics.

Uribe, M. (2017). "Is the Monetarist Arithmetic Unpleasant?" NBER Working Paper 22866.

Werning, I. (2024). "Recalculating Sargent and Wallace's 'Some Unpleasant Monetarist Arithmetic'." *Federal Reserve Bank of Minneapolis Quarterly Review* 44(3), 2–19.